

Process Capability – Waiting Time vs. Target

Assumptions

- Customer “specification”: checkout waiting time should be less than or equal to 7 minutes.
- No lower limit; shorter is better as long as service is correct.
- From baseline data: mean waiting time = 8.5 minutes, standard deviation = 2.3 minutes.

Simple Capability View

Item	Value
Mean (μ)	8.5 min
Standard deviation (σ)	2.3 min
Upper spec limit (USL)	7.0 min
Approx. Z-score at USL	$(7.0 - 8.5) / 2.3 \approx -0.65$

Plain-Language Interpretation

- The average wait (8.5 minutes) is already above the informal target of 7 minutes.
- Many customers naturally fall beyond that target, so current “capability” is poor.
- Before doing a fancy Cp/Cpk calculation, it is clear that the process needs to be centered and tightened.

For this simple project, this basic capability view is enough to justify moving into Analyze and then Improve.